Calculated Fields, Parameters & LOD Expressions Assignment

Theoretical Questions:

Question 1: What is a Calculated Field and why is it important in business dashboards?

Answer

Calculating fields Create new data fields using formulas, conditions or aggregations.

Its also help to performing advanced calculation and creating customs metrics dynamically.

Steps why imp for business dashboards

- Apply logical condition
- Perform custom calculations.
- Enable advanced analysis using mathematical and logical function.
- It's important because it allows analysts to derive new insights (e.g., profit margin, delivery classification) without altering the raw dataset.
- In dashboards, calculated fields help answer specific business questions like efficiency, profitability, or customer segmentation.

Question 2: Why can't parameters change visuals directly in Tableau?

Answer

Parameter is statics single value input (like dropdown slicer).

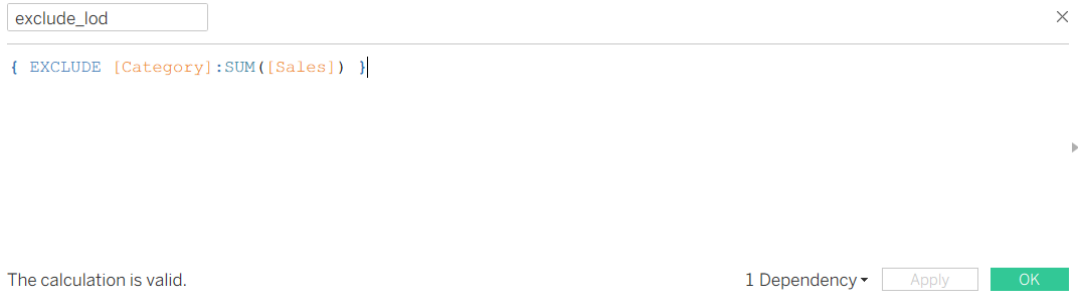
They don't directly change visual they are not tied to data instead, they must be combined with calculated fields or filter.

Example: A parameter "Top N product" only works when linked to a calculated field that ranked products.

Question 3 : When should EXCLUDE LOD be used?

Answer

Level of detail to allowed LOD Expressions allow you to control at what level Tableau should perform a calculation, independent of the visualization level.



Question 4: Difference between FIXED and INCLUDE LOD expressions?

Answer

LOD		
	Fixed	**Include**
Meaning	Fixed locks the calculation at a specific level	INCLUDE adds extra detail to the calculation
	It ignores the dimensions in the view	Calculation happens at a more detailed level
	Mostly ignores filters (except context filters)	Result is then aggregated back to the view
Syntax	{ FIXED [Dimension] : AGG([Measure]) }	{ INCLUDE [Dimension] : AGG([Measure]) }

Task 1: Delivery Performance Classification (Calculated Field)

Business Need: Operations wants to track delivery efficiency.

Task: Classify orders as Fast or Delayed.

Condition

- Delivery ≤ 30 min → Fast
- Else → Delayed

Answer

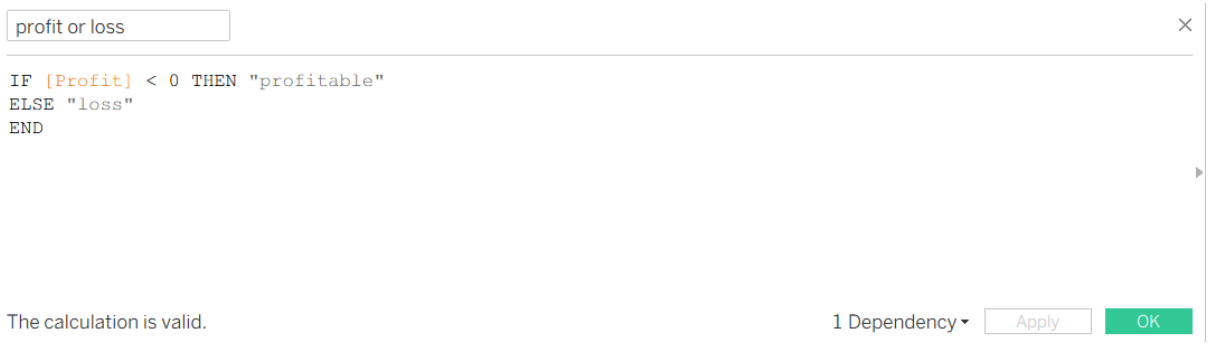


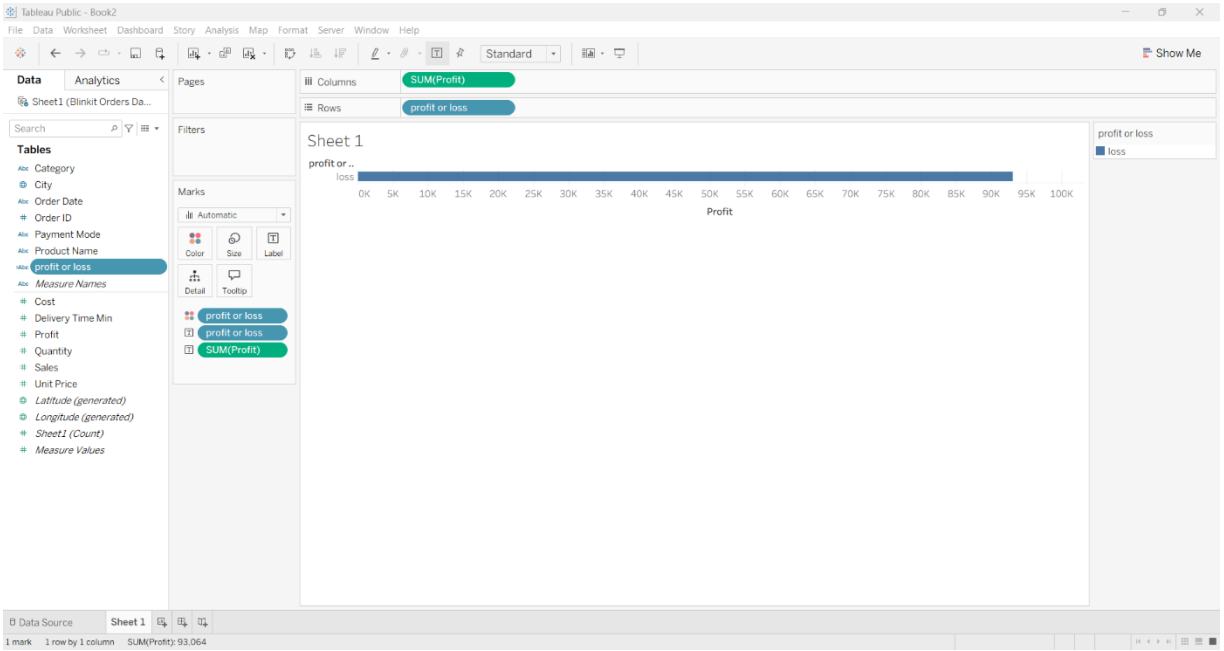
Task 2: Loss-Making Orders Identification

Business Need: High sales do not always mean high profit.

Task: Identify orders resulting in loss.

Answer



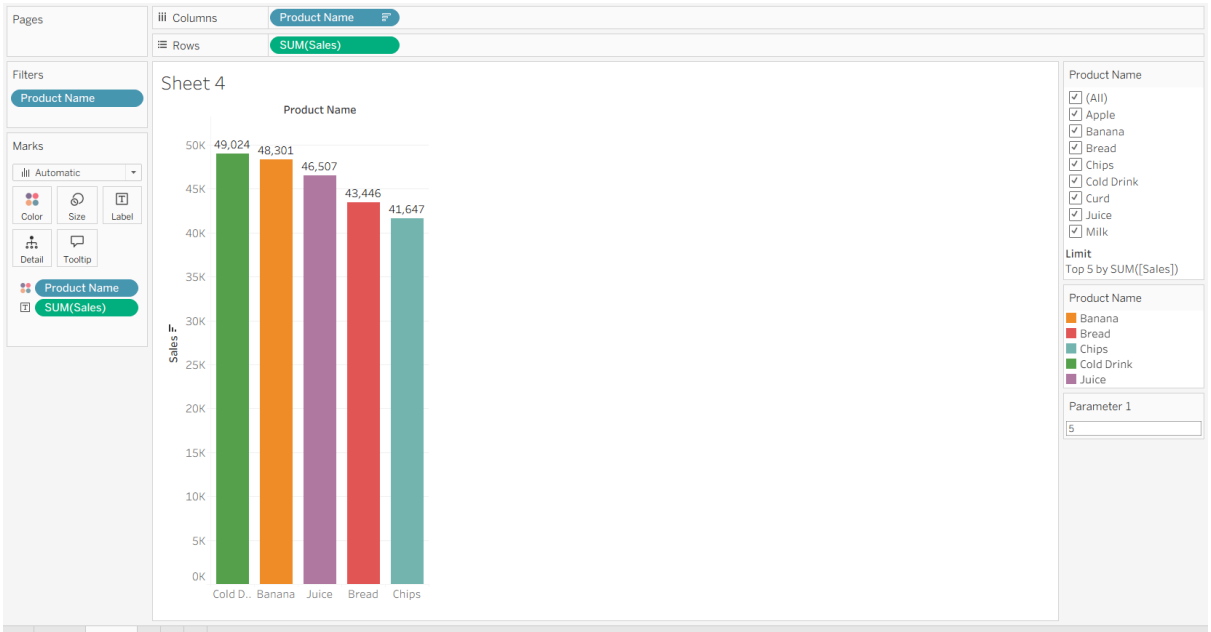


Task 3: Top N Products Analysis (Parameter)

Business Need: The product team wants to focus on top-performing products.

Task: Create a Top N Products dynamic analysis.

Answer

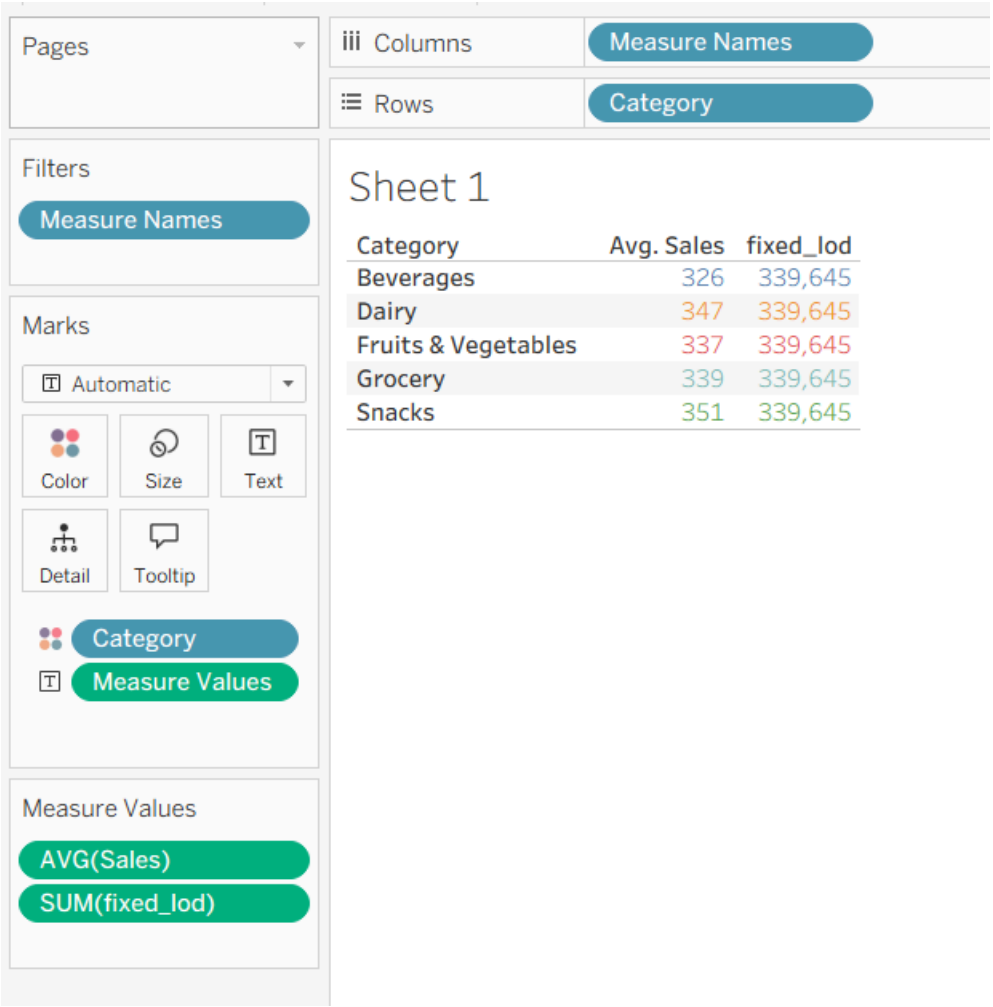


Task 4: City-Level Benchmarking (FIXED LOD)

Business Need: Compare city performance independent of category filters.

Task: Calculate average sales per city using FIXED LOD.

Answer

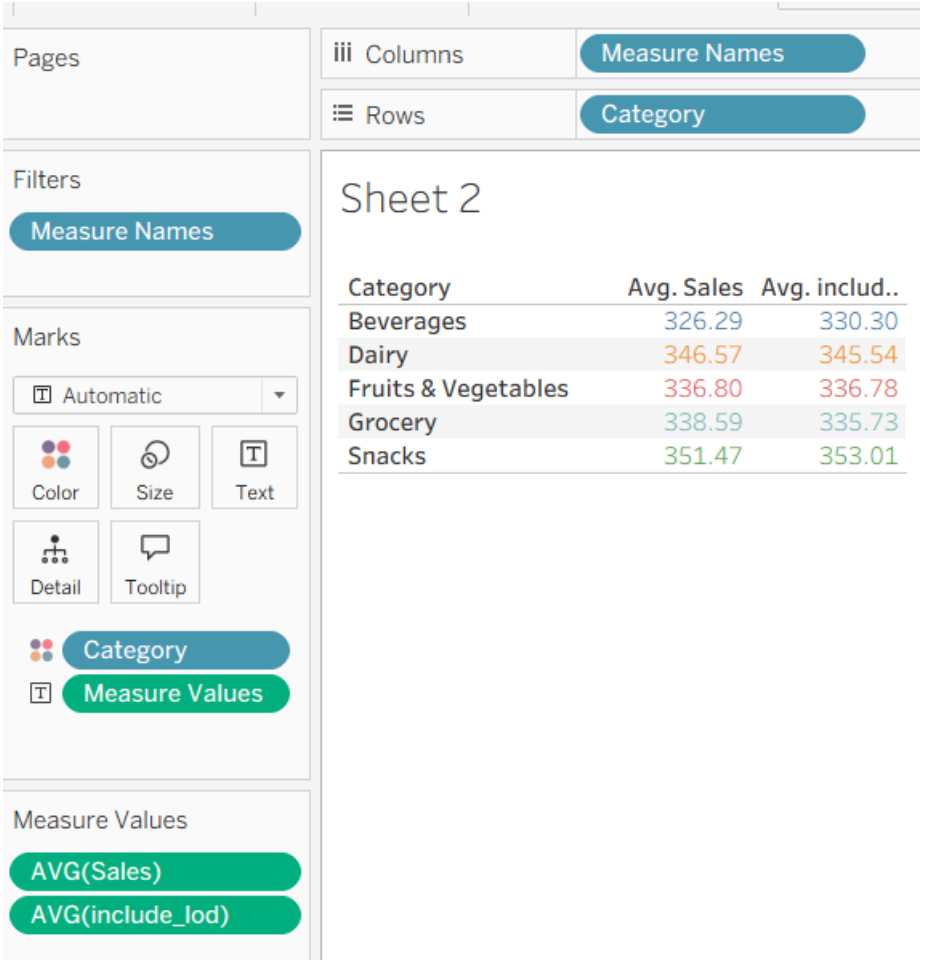


Task 5: Customer/Product Detail Impact (INCLUDE LOD)

Business Need: The sales team wants average sales considering product-level detail, even when product is not shown.

Task: Create a calculation that finds the average sales at the product level, but allows the result to be used at a higher level (such as Category or Region). Use INCLUDE LOD.

Answer


Pages
Columns
Measure Names
Rows
Category
Filters
Measure Names
Marks
Automatic
Color
Size
Text
Detail
Tooltip
Category
Measure Values
Measure Values
AVG(Sales)
AVG(include_lod)
Sheet 2
Category
Avg. Sales
Avg. includ..
Beverages
326.29
330.30
Dairy
346.57
345.54
Fruits & Vegetables
336.80
336.78
Grocery
338.59
335.73
Snacks
351.47
353.01